

Amphiphilic Dicyclopentenyl/Carboxybetaine-Containing Copolymers for Marine Fouling-Release Applications

Florian Koschitzki, Robin Wanka, Lennart Sobota, Julian Koc, Harrison Gardner, Kelli Z. Hunsucker, Geoffrey W. Swain, and Axel Rosenhahn*



Cite This: *ACS Appl. Mater. Interfaces* 2020, 12, 34148–34160



Read Online

ACCESS |



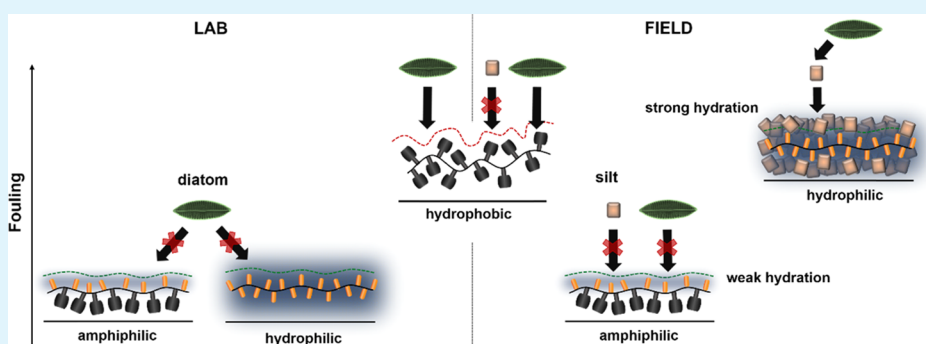
Metrics & More



Article Recommendations



Supporting Information



ABSTRACT: Zwitterionic materials received great attention in recent studies due to their high antifouling potential, though their application in practical coatings is still challenging. Amphiphilic polymers have been proven to be an effective method to combat fouling in the marine environment. This study reports the incorporation of small amounts of zwitterionic carboxybetaine methacrylate (CBMA) into hydrophobic ethylene glycol dicyclopentenyl ether acrylate (DCPEA). A new set of copolymers with varying amphiphilicities was synthesized and coated on chemically modified glass substrates. The antifouling capabilities were assessed against the diatom *Navicula perminuta* and multiple species in the field. Unsurprisingly, high diatom densities were observed on the hydrophobic control coatings. The integration of small zwitterionic contents of only ~5 wt % was already sufficient to rapidly form a hydrophilic interface that led to a strong reduction of fouling. Ultralow fouling was also observed for the pure zwitterionic coatings in laboratory experiments, but it failed when tested in the real ocean environment. We noticed that the ability to absorb large amounts of water and the diffuse nature of the interphase correlate with the adsorption of silt, which can mask the hydrophilic chemistries and facilitate the settlement of organisms. The amphiphilic coatings showed low fouling in dynamic short-term field exposures, which could be explained by the reduced tendency of the coatings for sediment adsorption.

KEYWORDS: antifouling, biofouling, zwitterion, diatom, methacrylate, *Navicula perminuta*

INTRODUCTION

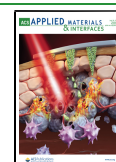
Prevention of the accumulation of biomass on submerged surfaces involves great ecological and economic value. The higher roughness of hard paint coatings on ship hulls due to biofouling in the marine sector directly correlates with higher frictional drag and increased fuel consumption.^{1,2} In addition, ships play a huge role in the transport mechanism of inadvertent carryover of nonindigenous, invasive species to international waters, leading to ecological damage.^{3,4} These factors lead to a demand for antifouling (AF) and fouling-release (FR) coatings that either inhibit the settlement or facilitate the detachment of marine micro- and macrofoulers.⁵ As the environmental issues of biocide-containing AF coatings (e.g., tributyl tin-based self-polishing copolymers) became apparent, the research on alternative AF strategies and coating technologies intensified.^{6,7} The formation, stability, and release of fouling are essentially influenced by the surface

chemistry. Baier demonstrated that the relative attachment of fouling organisms is linked to surface energy.⁸ Systematic studies on self-assembled monolayers (SAMs) revealed the relevance of steric repulsion, charge neutrality, and hydration for the design of low-fouling coatings.^{9,10} In recent years, in particular, amphiphilic coatings have received increasing attention. The combination of hydrophilic components in the direct vicinity of hydrophobic polymers provides enhanced fouling-release properties.^{11–17} Although the precise mechanism is still an ongoing study, clearly, the rearrangement of

Received: April 24, 2020

Accepted: June 22, 2020

Published: June 22, 2020



the polymer under water and the associated hydration of the hydrophilic groups are advantageous for the removal of organisms under shear stress.^{11–18} Gudipati et al. described increased fouling resistance of hyperbranched fluoropolymers (HBFP)—poly(ethylene glycol) (PEG) coatings on biomacromolecules and *Ulva* sporelings in comparison to a poly(dimethylsiloxane) (PDMS) elastomeric standard.¹⁸ Krishnan et al. blended a surface-active fluoropolymer with PEG to observe an increased removal of settled diatoms.¹⁵ Similar effects were obtained for PEGylated-fluorinated polystyrene by Martinelli et al. However, the AF and FR properties differed distinctly from *Ulva linza* to *Navicula perminuta* depending on the respective composition of the amphiphilic styrene polymer.¹⁹ Amphiphilic alginates with fluorinated side chains showed settlement reduction of green algae and diatoms.^{11,20} Also, amphiphilic polysiloxane coatings proved to have potential toward the resistance of marine organisms.²¹ While ethylene glycols are an established hydrophilic compound in amphiphilic systems, zwitterionic materials have drawn increasing attention due to their high resistance against nonspecific protein adsorption^{22–26} and freshwater and marine fouling.^{27–31} The hydrophobic compounds are usually restricted to butyl methacrylate, PDMS, and polystyrene. More recently, the Alexander group used combinatorial screening to test methacrylate libraries regarding their resistance to freshwater pathogens for biomedical applications.³² Among the discovered compounds was a combination of hydrophobic dicyclopentenyl- and hydrophilic ethylene glycol-containing methacrylate monomers.

A majority of the studies on amphiphilic polymers used poly(ethylene glycol) as the hydrophilic compound. Here, zwitterionic compounds could be useful alternatives. The first zwitterionic molecules incorporated into polymers were phosphatidylcholines, bioinspired by the AF properties of blood cell membranes.^{33,34} In reports of Zhang et al., sulfobetaine (SBMA) and carboxybetaine methacrylate (CBMA) monomers were immobilized on glass slides using atom transfer radical polymerization (ATRP), yielding ultra-low fouling polymer brushes with high resistance against nonspecific protein adsorption comparably to PEGylated surfaces.³⁵ The striking properties of polyzwitterions in respect of fouling were underlined in numerous studies. Jiang and co-workers demonstrated excellent fouling resistance of poly(carboxybetaine acrylamide) in undiluted human plasma.³⁶ In addition, zwitterionic polymer coatings drastically reduced the biofilm formation of *Pseudomonas aeruginosa*,^{27,37} settlement of barnacle cyprids,²⁸ attachment of spores of *Ulva*, diatoms *N. perminuta*,³⁸ and cell adhesion.³⁹ To increase the structural diversity of polyzwitterionic polymers and to investigate the effect of charge arrangement in the polymers, a range of sulfa- and sulfobetaines were developed that revealed the relevance of the chemistry of the zwitterionic group itself but also of the polymer backbone on their marine biofouling performance.^{40,41} When applied in the field, Koc et al. detected the incorporation of diatomaceous soil and inorganic material (silt) on zwitterionic coatings with a strong hydrogel character that subsequently enabled marine organisms to attach. This led to the idea that embedding of zwitterionic moieties in amphiphilic copolymers could reduce silt uptake and thus lead to optimal performance in the field.⁴² Previous studies have indicated already the good protein resistance of copolymerized carboxybetaine butyl methacrylate

on protein resistance⁴³ and lauryl methacrylate-based sulfobetaine and carboxybetaine coatings toward *Navicula incerta* even at small zwitterionic contents.⁴⁴

In this work, we created an amphiphilic methacrylate coating based on dicyclopentenyl as the hydrophobic compound³² and carboxybetaine as the hydrophilic building block. Commonly, when new coating chemistries are explored, polymers are copolymerized in solution, characterized, and then applied to the surface by, e.g., dip/spin-coating and either attached to the surfaces via van der Waals forces or thermal or photoinitiated tethered.^{40,43,44} If suitable solvents are used, monomer solutions can also be applied and polymerized directly on surfaces; however, this process demands very controlled conditions to prevent quick solvent evaporation and potential subsequent phase separation. Here, we report a facile method to polymerize amphiphilic monomers directly in situ at the surface under controlled conditions that do not suffer phase segregation or rough surface coatings. Ethylene glycol dicyclopentenyl ether acrylate (DCPEA) and carboxybetaine methacrylate (CBMA), with small amounts of ethylene glycol diacrylate (EGDA) as cross-linkers, were copolymerized in a sealed environment using a photoinitiator. To achieve a covalent connection to the glass substrates, they were functionalized with 3-(trimethoxysilyl)propyl methacrylate (MPTMS) as polymerizable groups. Free-radical-polymerized, amphiphilic methacrylates in various mixture ratios were characterized and studied with respect to their resistance against the marine biofouling organism *N. perminuta* and multiple species in the field.

■ EXPERIMENTAL SECTION

Chemicals. *N,N'*-Dimethylaminoethyl methacrylate (DMAEM), ethylene glycol dicyclopentenyl ether acrylate (DCPEA), ethylene glycol diacrylate (EGDA), 2,2-dimethoxy-1,2-diphenylethan-1-one (DMPAP), and 3-(trimethoxysilyl)propyl methacrylate (MPTMS) were purchased from Sigma-Aldrich. Ethanol, acetonitrile, and diethyl ether were purchased from VWR International. 2,2,2-Trifluoroethanol (TFE) was acquired from Roth. Acetone and ethylacetate were acquired from Fisher chemicals, and β -propiolactone was purchased from ABCR. All chemicals were used as received.

Synthesis of Carboxybetaine Methacrylate (CBMA). The carboxybetaine methacrylate 3-[[2-(methacryloyloxy)ethyl]-dimethylammonio]propionate was synthesized by the reaction of β -propiolactone and DMAEM following a previously published protocol (Zhang et al.).³⁵ In brief, DMAEM (800 μ L, 4.75 mmol) was dissolved in 25 mL of dried acetone. β -Propiolactone (358.0 μ L, 5.70 mmol) in 5 mL of dried acetone was added dropwise to the solution under nitrogen. The reaction mixture was cooled down to 4 °C and stirred at this temperature for 4.5 h. The resulting white precipitate was centrifuged at 5000 rpm in Falcon tubes (Hettich centrifuge universal 320R) and then washed with anhydrous ether. The product was dried under reduced pressure and a white solid was obtained (yield 62%). The monomer was stored at 4 °C under argon. Nuclear magnetic resonance (NMR) was measured in deuterated water as a solvent. ¹H NMR (200 MHz, D₂O): δ 1.97 (s, 3H, =CCH₃), 2.76 (t, 2H, CH₂COO), 3.22 (s, 6H, NCH₃), 3.70 (t, 2H, NCH₂), 3.82 (t, 2H, CH₂N), 4.66 (t, 2H, OCH₂), 5.80 (s, 1H, =CH), 6.19 (s, 1H, =CH).

Nuclear Magnetic Resonance. Nuclear magnetic resonance (NMR) characterization was carried out on a Bruker Avance DPX-200 spectrometer at 303 K equipped with a Bruker B-ACS 60 sample changer. The data was analyzed using software MestReNova 10.0.2. Chemical shifts δ are given in parts per million (ppm) against the internal standard. The following abbreviations were used: s (singlet), d (doublet), t (triplet), dd (doublet of doublet), dt (doublet of

triplet), td (triplet of doublet), and m (multiplet). The coupling constants J are given in hertz (Hz).

Preparation of Dodecanethiol (DDT) SAM. As a reference, dodecanethiol (DDT) was prepared as a self-assembled monolayer (SAM) on gold-coated microscope slides according to a previously published protocol.⁴⁵ Briefly, the slide was cleaned under a UV lamp for 1 h, sonicated in ethanol for 3 min, rinsed with ethanol, and dried under a stream of nitrogen before immersion into a 1 mM thiol solution for 24 h. The DDT SAM was again sonicated for 3 min and rinsed with ethanol. After drying with nitrogen, the samples were stored under argon until used.

Surface Preparation. Prior to the preparation of the polymer coatings, smooth glass slides (Nexterion B, Schott, Germany) were modified with a methacrylate tie coat, with slight modifications to a previous reported procedure.⁴⁷ In brief, 20 min of oxygen plasma (0.4 mbar, 80 W, 22 kHz, miniFlecto-PC-MFC, Gala Instruments, Germany) was applied to activate the surfaces by the generation of hydroxy groups. To couple 3-(trimethoxysilyl)propyl methacrylate, a solution of 10% (v/v) of the silane in toluene was degassed for an hour before immersing the plasma-treated glass slides. The reaction mixture was sonicated at 50 °C overnight. After 16 h, the glass slides were rinsed with ethanol, dried in a stream of N₂ gas, and cured in a vacuum oven at 85 °C for 30 min. To remove unbound MPTMS, the substrate was sonicated for 20 s in ethanol, before being dried in a stream of nitrogen. The slides were used immediately after preparation or stored under argon.

Preparation of Polymer Coatings. Polymer coatings were prepared via photoinitiated free-radical polymerization. As both the monomers and the surface carried polymerizable groups, the mechanism is probably a mixture of “grafting to” and “grafting from”. Solutions of carboxybetaine methacrylate (CBMA) and ethylene glycol dicyclopentenyl ether acrylate (DCPEA) in trifluoroethanol (TFE) were prepared in different weight ratios (0, 5, 10, 15, and 100% (w/w) CBMA/DCPEA). Ethylene glycol diacrylate (EGDA, 3% (w/w)) as the cross-linker and 2,2-dimethoxy-1,2-diphenylethane-1-one (DMPAP, 1% (w/w)) as the photoinitiator dissolved in acetonitrile were added before degassing the light-protected solution for 45 min. As control coatings, a mixture of carboxybetaine methacrylate and butyl methacrylate was prepared with the same protocol using weight ratios of 0 and 5% (w/w) (CBMA/butyl methacrylate (BMA)). To exclude oxygen and avoid evaporation of the solvent, radical polymerization was conducted in an airtight system applying a HybriWell sealing system (SecureSeal adhesive chamber, chamber volume 275–320 μL, size $L \times W \times$ thickness 60 mm $\times$ 21 mm $\times$ 0.15 mm, ports diam. 3.2 mm; Grace Bio-Labs) to the MPTMS-modified glass substrate.⁴⁷ Depending on the content of carboxybetaine, the polymerization was initiated between 5.5 and 7.5 min (0 wt %: 5.5 min, 5 wt %: 6.0 min, 10 wt %: 6.3 min, 15 wt %: 6.5 min, 100 wt %: 7.5 min) by UV irradiation (UVAcube 100, Dr. Hönle AG, 100 W iron-doped mercury vapor lamp, cutoff filter >310 nm, source–sample distance 18 cm, 15 mW/cm²). The reaction chamber was kept sealed for a further 16 h to allow the remaining monomers to diffuse and react. After opening the reaction chamber, the obtained polymers were leached in TFE for 30 min and rinsed with water to remove unpolymerized monomers. The coated slides were dried in a stream of nitrogen and stored under argon until tested.

Contact Angle Goniometry. The static water contact angles (WCAs) were determined using a custom-built goniometer. A charge-coupled device (CCD) camera recorded a droplet of tridistilled water (~50 μL) deposited on the surface. To obtain the contact angle, the droplet shape was fitted with Young's equation. Under water, contact angles were obtained with a separate custom-built goniometer. A captive air bubble is recorded with a CCD camera on the surfaces immersed in a transparent water container with the coating facing down. Contact angles were manually determined using the angle tool of the software package ImageJ. The drawn angle was manually adjusted to fit the surface of the substrate and the other side to describe the tangent of the air bubble to obtain the exterior angle. For each sample, three positions on the

surface were measured. Reported values are the average of three measurements. Error bars imply the standard deviation.

Attenuated Total Reflection (ATR)-Fourier Transform Infrared (FTIR) Spectroscopy. Total internal reflection Fourier transform infrared (ATR-FTIR) spectra were measured with a Vertex 70 spectrometer (Bruker, Ettlingen, Germany) equipped with a platinum ATR diamond crystal. The shown spectra display the average of 50 scans at a resolution of 4 cm⁻¹ in a range of 650–4000 cm⁻¹. The FTIR was recorded by an RT-DLaTGS detector.

Stability and Water Uptake. To determine the stability and swelling of the coatings, they were incubated for different durations in Milli-Q (MQ) or saltwater (SW) consisting of the seven most abundant components (>50 ppm) of seawater.⁴⁸ SW was preferred to artificial seawater (ASW) to prevent organic deposits.⁴⁹ After different immersion times, the coatings were retrieved, gently wiped to get rid of excess water drops on the glass slide, and the weight (M_{wet}) was determined on a Mettler Toledo XA105DU high-precision scale (Gießen, Germany). After drying for 15 min at 65 °C, the weight was measured again (M_{dry}). The uptake of water was calculated after 144 h of immersion as the relative mass increase due to the presence of water in the film with the equation, relative water uptake = $\frac{M_{\text{wet}} - M_{\text{dry}}}{M_{\text{dry}}}$. To reveal the degradation of the material in the

corresponding medium after different immersion times, the dry weight after immersion (M_{dry}) was compared to the one of the pristine coatings M_0 prior to immersion and calculated with the equation, relative change of mass = $\frac{M_{\text{dry}}}{M_0}$. Each value represents the mean of three replicates, with error bars as the standard error.

Atomic Force Microscopy (AFM). Atomic force microscopy (AFM) images were obtained on a NanoWizard 3 (JPK, Germany) using a MULTI75GD-G (Budget Sensors, Bulgaria) with a 3 N/m force constant for a scan range of 20 $\times$ 20 μm² (4 V target amplitude, 2 V setpoint, 750 Hz i-gain, 89 μm/s tip velocity) in air and under SW. The coating roughness and the lateral deflection of the cantilever were determined utilizing the root-mean-square (RMS) roughness option of Gwyddion on plane fitted images. The presented values depict the mean of three areas on the height trace image, while error bars represent the standard error.

Dynamic Attachment Assay of *N. perminuta*. The dynamic attachment assay in a microfluidic environment was carried out following a previously published protocol.⁵⁰ The diatom *N. perminuta* was purchased from the Culture Collection of Marine Algae and Protozoa (Dunbeg, Great Britain) and transferred to sterilized seawater from SeaPure (CaribSea, Fort Pierce, Florida) with Guillard f/2 and silica culture medium at a pH of 8.1. Diatoms were cultured in a light cabinet (Snijder Scientific, Tilburg, Netherlands) using a day/night light/dark cycle of 12/12 h. For the experiments, the culture medium was removed and replaced by nutrient-free ASW medium. The settled cells were removed from the bottom of the flasks by sonication and diluted to the required concentration, which was determined in a UV/vis spectrometer Cary 60 (Agilent Technologies, Waldbronn, Germany) at a wavelength of 444 nm (OD₄₄₄). For the assay, a diatom suspension of 2 million diatoms/mL was rinsed through microfluidic channels that contained the coatings to be tested for 90 min at a laminar flow exerting a constant wall shear stress of 0.16 Pa. Subsequently, the channels were rinsed with pure seawater at the same flow speed for 10 min to remove unattached cells from the channels.

Rotating Disk Field Assay. To test the samples in the real marine environment, they were mounted on a custom-made rotating disk at Port Canaveral, Florida, previously described in detail.⁵¹ Briefly, the samples were mounted flush to the surface of the disk and a rotation speed of 12 rpm was applied to the disk to allow a controlled wall shear stress of 0.18 Pa to establish at its surface. After 5 days of dynamic immersion, the samples were retrieved from the ocean, immersed into a solution of 2.5 wt % glutaraldehyde (Fisher Scientific, Pennsylvania) for 10 min, and washed with MQ water prior to drying and further analysis.

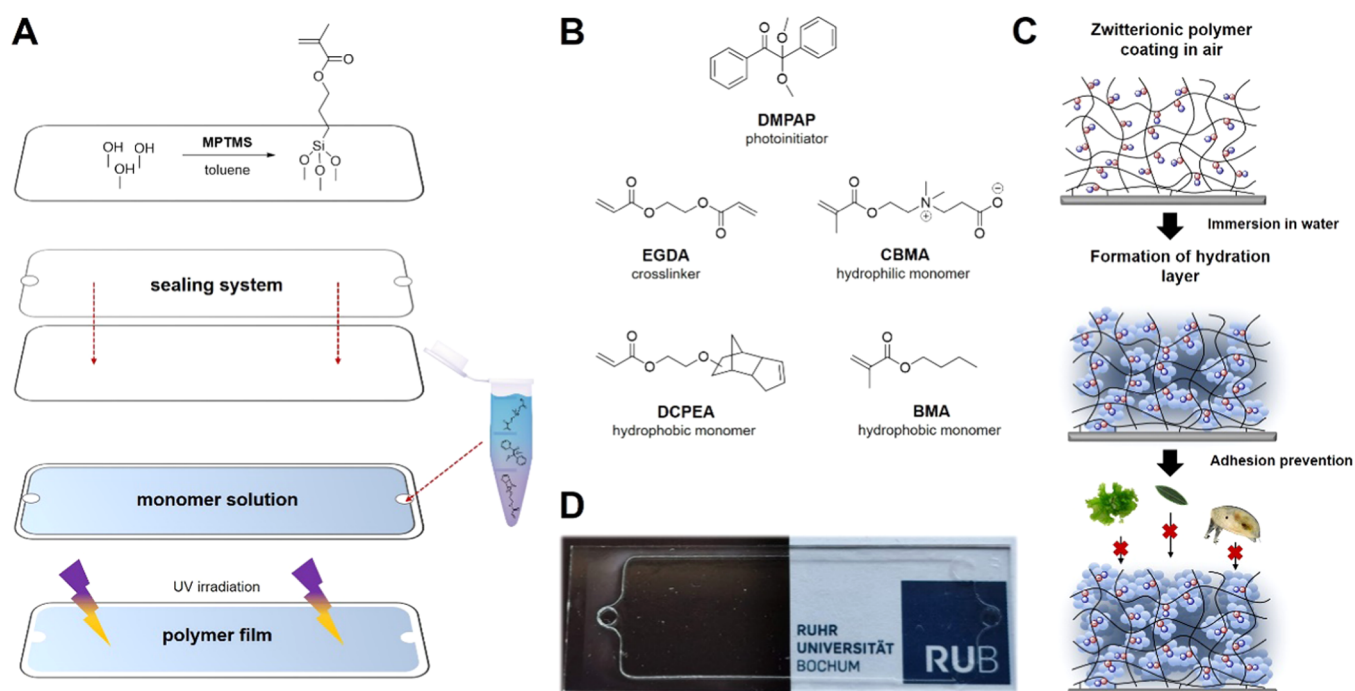


Figure 1. (A) Schematic representation of the preparation of the amphiphilic polymer coatings: a glass substrate is functionalized with MPTMS as the tie coat. A sealing system is applied to the slide and filled with monomer solution prior to UV irradiation. (B) Structures and abbreviations of the amphiphilic polymer coatings: main building blocks consist of ethylene glycol dicyclopentenyl ether acrylate (DCPEA) and methacrylate 3-[[2-(methacryloyloxy)ethyl]dimethylammonio]propionate (CBMA). Small amounts (3.0 wt %) of ethylene glycol diacrylate (EGDA) as cross-linkers and 1.0 wt % of 2,2-dimethoxy-1,2-diphenylethan-1-one (DMPAP) as the photoinitiator were used. (C) Illustration of the behavior of the amphiphilic coatings: zwitterionic polymers form a hydration layer in water that prevents fouling. (D) Exemplary image of polymer coating on a glass substrate.

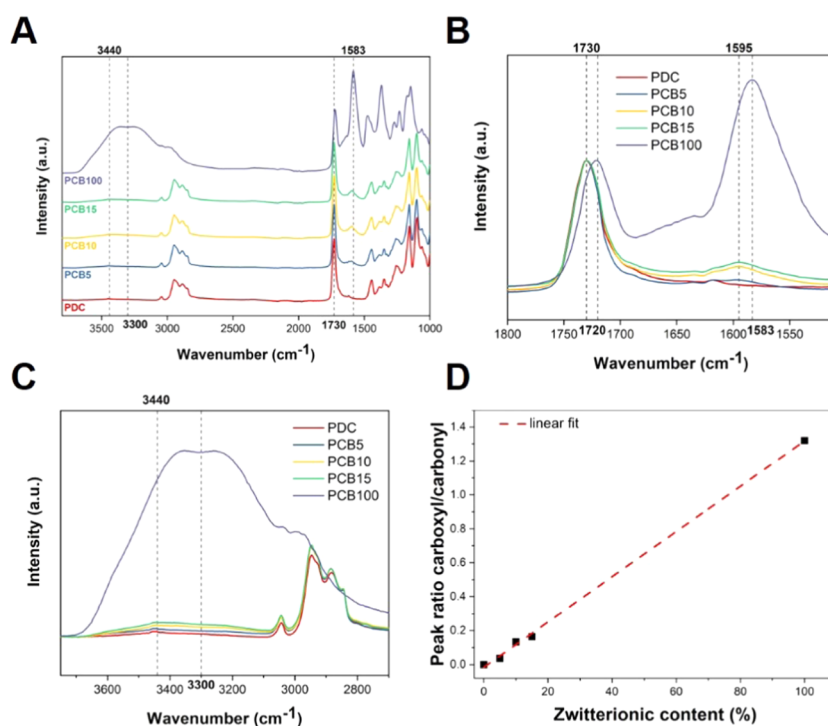


Figure 2. (A) ATR-FTIR spectra of the synthesized polymer coatings. (B) Magnified ATR-FTIR spectra of the polymer coatings in the section of the C=O ester band (1730 cm^{-1}) and the C=O carboxyl band (1583 cm^{-1}). (C) Magnified ATR-FTIR spectra of the polymer coatings in the section of the OH-stretching vibration (3300 cm^{-1}). (D) Peak ratio of the C=O carboxyl band (1583 cm^{-1}) to the C=O carbonyl band (1730 cm^{-1}) plotted against the zwitterionic content of CBMA in the monomer solution used to prepare PDC, PCB5, PCB10, PCB15, and PCB100.

Table 1. Polymer Composition and Water Contact Angle of the Coatings

	monomer unit composition			uv irradiation time (min)	curing time (h)	contact angle	
	CBMA/DCPEA (wt %/wt %)	cross-linker (wt %)	initiator (wt %)			in air (deg)	under water (deg)
PDC	0/100	3	1	5.5	16	99 ± 1	58 ± 3
PCB5	5/95	3	1	6.0	16	96 ± 1	18 ± 2
PCB10	10/90	3	1	6.3	16	93 ± 3	15 ± 2
PCB15	15/85	3	1	6.5	16	88 ± 3	14 ± 2
PCB100	100/0	3	1	7.5	16	42 ± 2	9 ± 1

Microscopy and Image Analysis. For quantification of the density of attached diatoms after the microfluidic accumulation assay and after field exposure, the samples were imaged by phase-contrast and fluorescence microscopy. An automated phase-contrast Nikon Eclipse Ti microscope (10× phase-contrast objective CFI Plan Fluor DLL NA 0.3, Nikon, Germany) with a BV-2A filter set automatically recorded images on a predefined grid on the surface to produce 30 representative fields of view (dynamic attachment assay) or 60 fields of view (rotating disk field assay) using the automated focusing option of NIS Elements AR 4.30.02 (Nikon, Germany). The software was further used to automatically count the density of diatoms on the samples after the microfluidic assay. In the case of clustered cells on the sample, counting was done manually. For the field assay, the attachment was analyzed through fluorescence coverage, detecting the autofluorescence of the organisms. Differences between distinct chemistries were tested for their statistical significance using an analysis of variance (ANOVA) with post hoc Tukey analysis (significance level $\alpha = 0.05$).

Sand Immersion. The uptake of sand was determined using natural sand sediment extracted from the sea bottom at Port Canaveral (Florida), where the rotating disk field assay was performed. After autoclaving, the sand was dissolved in sterile filtered seawater (CaribSea Pure, Fort Pierce, Florida) at a concentration of 37.5 g/L. Polymer samples were immersed in the sand suspension for 10 min or 60 min on a linear shaker at 60 rpm, air-dried, and analyzed by UV/vis spectroscopy (Agilent, Cary 60 UV–vis, Germany).

RESULTS

Sample Preparation and Characterization. A facile approach to copolymerize amphiphilic monomers directly in contact with a preactivated surface by photoinduced free-radical polymerization was used (Figure 1A).^{46,47} Methacrylate-terminated silanes were used as covalent anchor groups to allow covalent linkage of the polymer to the substrates. To avoid oxygen diffusion and ensure controlled polymerization conditions without solvent evaporation, an airtight sealing system was applied to the substrate. The monomer solution contained additionally 2,2-dimethoxy-1,2-diphenylethan-1-one (DMPAP) as the photoinitiator and ethylene glycol diacrylate (EGDA) as the cross-linker (Figure 1B). To comply with the solubilities of the different monomers, 2,2,2-trifluoroethanol (TFE) was used as a solvent. The monomers were finally free-radical polymerized by UV initiation. The irradiation time depended on the monomer ratio as the hydrophilic methacrylate compound required slightly longer curing times than the hydrophobic acrylate. The resulting polymer coatings are described as PBMA (100 wt % BMA), PBMA-CB5 (95 wt % BMA, 5 wt % CBMA), PDC (100 wt % DCPEA), PCB5 (95 wt % DCPEA, 5 wt % CBMA), PCB10 (90 wt % DCPEA, 10 wt % CBMA), PCB15 (85 wt % DCPEA, 15 wt % CBMA), and PCB100 (100 wt % CBMA).

To verify the incorporation of hydrophilic zwitterionic units (CBMA) into the hydrophobic (DCPEA)-containing polymer, attenuated total reflection Fourier transform infrared spec-

troscopy (ATR-FTIR) and static water contact angle (WCA) goniometry were used. As shown in Figure 2, the IR spectra of the polymer coatings are in general very similar, but two main changes become apparent when the content of CBMA is increased. The broad band at around 3300 cm^{-1} represents the OH-stretching vibration of the carboxylic acid (OH) of the zwitterion, which is translocated from 3450 cm^{-1} in PCB5 to 3300 cm^{-1} in PCB100 (Figure 2C). The ester (C=O) absorption band at 1730 cm^{-1} is visible in DCPEA and CBMA, while the carboxyl (C=O) absorption at 1583 cm^{-1} solely belongs to CBMA and verifies the successful embedment of varying amounts of zwitterionic material. Interestingly, both C=O peaks are shifted to lower wavenumbers with increasing zwitterionic content in Figure 2B (ester band from 1730 cm^{-1} in PDC to 1720 cm^{-1} in PCB100, carboxyl band from 1595 cm^{-1} in PCB5 to 1583 cm^{-1} in PCB100). The intensity ratio of these two characteristic peaks can be used as an indication of the hydrophilic content in the amphiphilic polymer. The carbonyl peak (1730 cm^{-1}), which is present in DCPEA and CBMA, is compared to the carboxyl peak (1583 cm^{-1}), which solely represents the content of CBMA. Compared to the homopolymers PCB100 and PDC, the carboxyl signal increases with increasing CBMA content (Figure 2A,B). The corresponding analysis Figure 2D shows that the CBMA content in the films as measured by ATR-FTIR was linear (slope = 1.13, $R^2 = 0.99$) with its content in the monomer solution. This indicates that both compounds, CBMA and DCPEA, were successfully copolymerized.

The wettability of the coatings was characterized by static water contact angle (WCA) goniometry (Table 1). For the hydrophobic PDC reference, a WCA of $99 \pm 1^\circ$ was obtained, while the one on the hydrophilic PCB100 was $42 \pm 2^\circ$. The amphiphilic coatings of CBMA and DCPEA from PCB5 ($96 \pm 1^\circ$) over PCB10 ($93 \pm 3^\circ$) to PCB15 ($88 \pm 3^\circ$) showed decreasing WCAs with increasing zwitterionic content, which is in line with the expectations. It is interesting to note that the absolute values, even for PCB15 with 88° , are still relatively hydrophobic. For comparability reasons, the contact angle was measured immediately after the water droplet touched the surface. We noticed that if the polymer was allowed to reorient in the presence of water, the contact angle decreased. The determination of an equilibrium sessile droplet contact angle of a hydrated coating suffered from the fact that full hydration in air cannot be achieved. Therefore, we further assessed the reorganization via captive air bubbles under water contact angle goniometry (Figure 3). To demonstrate the comparability of the contact angles obtained by two different methods, a dodecanethiol (DDT) self-assembled monolayer (SAM) was included in the experiment. The hydrophobic control DDT generally experiences no reorientation under water and showed a constant WCA of $97 \pm 5^\circ$ for both the measurements in air and under water. However, for the

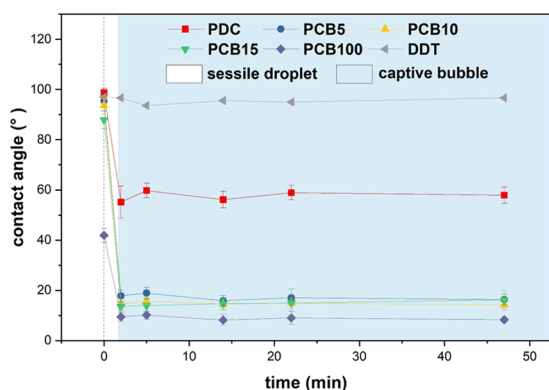


Figure 3. Change of contact angle as a function of time. The first measurement point represents a sessile water droplet in air. The following values were obtained by measuring a captive air bubble on polymer coatings immersed in water. A DDT reference that barely reorients under water was added to the experiment to demonstrate that the two different measurement methods yield the same results.

polymer coatings after an immersion time of only 2 min, a sharp decrease of the contact angles was observed. For PDC and PCB100, we noted a decrease of 30–40°, whereas the amphiphilic coatings even showed a decrease of nearly 80°, suggesting extensive reorientation under water. All amphiphilic and the pure zwitterionic coating reached contact angles <20°, which quickly equilibrated.

Prior to testing the antifouling and fouling-release properties, the stability of the polymer coatings was tested by immersion in saltwater (SW) containing the essential marine salts to simulate seawater conditions but avoiding the deposition of organic substances on the surface (as it occurs from artificial seawater).⁴⁹ After exposure to SW for up to 13 days, the samples were retrieved and measured on a high-precision scale to determine their stability. Figure 4A shows that all polymers retained their initial mass ($\geq 92\%$) and were stable for 13 days in SW. The stability was confirmed by FTIR spectroscopy as shown in Figure 4B, which reveals that the compositions of DCPEA and CBMA in the amphiphilic polymers (PCB5 exemplarily) did not show any major

changes due to the immersion. The ability to form stable hydration layers is one of the reasons for the inertness of zwitterions and plays a crucial role in the antifouling properties of zwitterion-containing coatings.⁵² In Figure 4C, the swelling capacity of immersed polymers after 6 days in MQ and SW is shown, which was analyzed by gravimetric analysis. As expected, the water uptake of the polymer coatings increased with increasing hydrophilicity, and hydrophobic PDC showed the weakest swelling (1% in MQ and 1% in SW) and PCB100 showed the strongest swelling (330% in MQ and 245% in SW). For the amphiphilic polymers PCB5 (3% in MQ and 3% in SW), PCB10 (8% in MQ and 7% in SW), and PCB15 (5% in MQ and 6% in SW), an increasing tendency to swell with higher zwitterionic content was observed, which is according to the expectations. The only exception is PCB15, for which slightly lower swelling than that for PCB10 was observed. No significant difference of the swelling in MQ compared to SW was found for the amphiphilic coatings and PDC, whereas for PCB100, surprisingly, the water uptake was significantly lower in SW than that in MQ. In view of the usually observed antipolyelectrolyte behavior of polyelectrolytes, higher swelling in SW than that in MQ water would have been expected. The introduction of salt allegedly breaks up the dipole–dipole attraction of zwitterionic moieties that usually form intra-group, intrachain, and interchain salt bonds,⁵³ thus allowing an expansion of the polymer chain and liberated zwitterionic units causing increased swelling.⁵⁴ In this case, CBMA seems to behave more as a typical polyelectrolyte. Overall, the swelling results followed the trend of the water contact angles, i.e., that more hydrophilic coatings are capable of absorbing larger amounts of water.

To provide information on the surface topography and surface roughness, 20 $\mu\text{m} \times 20 \mu\text{m}$ AFM images on all samples were recorded. AFM images were acquired under dry (Figure 5A–E) and wet (SW) (Figure 5F–J) conditions to gain insight into any possible morphological changes of the coating surface since all of the following biological assays were performed in water. Several protruded spots distributed over the coatings were observed for the hydrophobic PCD (Figure

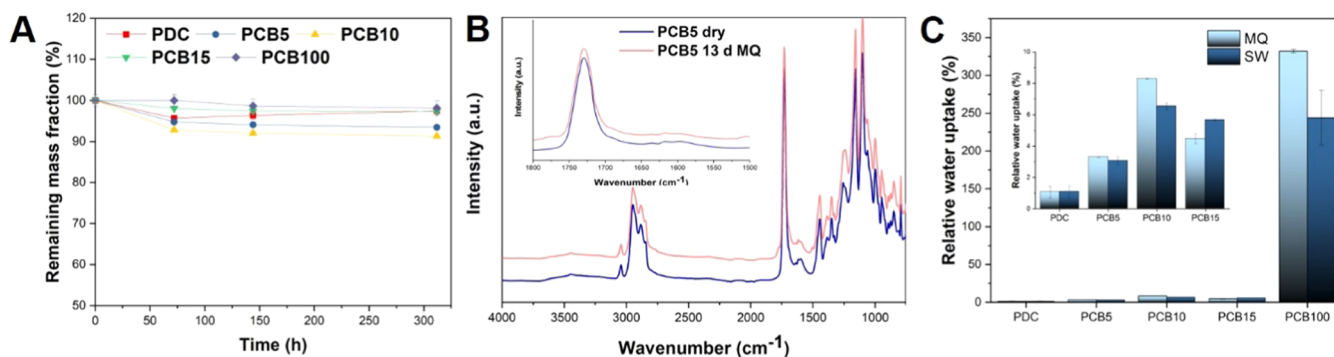


Figure 4. (A) Stability of the polymer coatings: relative change of mass prior to and after immersion in saltwater (SW, pH 8) for 72, 144, and 312 h at room temperature as determined by gravimetric analysis. After immersion, the coatings were dried and the mass was compared to the one of the dry pristine samples prior to immersion. Each value is the average of three independent measurements on different samples. Error bars represent the standard error of the mean. (B) Exemplary stacked ATR-FTIR spectra of PCB5 prior to and after immersion in Milli-Q (MQ) water for 312 h. The inset shows the region of the C=O ester and the carboxyl peak. (C) Relative mass increase due to swelling of the polymer coatings as determined by gravimetric analysis immediately after retrieval from 144 h immersion in MQ and SW and measured after 15 min in a vacuum drying oven at 65 °C. The water uptake is calculated as the relative mass gain of the wet polymers compared to the dry polymers. The inset shows a magnified view of the relative mass increase in the range of 0–10%. Each value represents the average of three independent measurements on different samples. Error bars are the standard error of the mean.

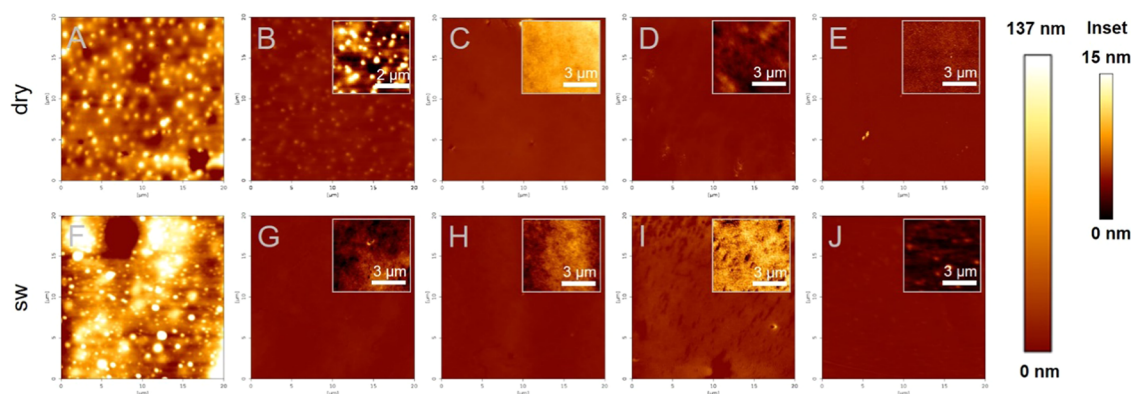


Figure 5. Representative AFM images ($20\ \mu\text{m} \times 20\ \mu\text{m}$) that characterize the morphology of the coatings and allow to determine the surface roughness. Top row: morphology in air for PCD (A), PCB5 (B), PCB10 (C), PCB15 (D), and PCB100 (E). Displayed in the bottom row are images recorded in SW: PCD (F), PCB5 (G), PCB10 (H), PCB15 (I), and PCB100 (J). The z-height is standardized to 137 nm for all AFM images. The z-height of the insets is standardized to 15 nm to illustrate the nanomorphology of the individual polymers.

SA). Similar features with a much smaller height were detected for PCB5 in Figure 5B. PCB10, PCB15, and the pure zwitterionic coating PCB100 were rather smooth and featureless. In SW, the polymer PCB5 also showed much smoother characteristics compared to its dry state, but a few structures that appeared like cracks became noticeable. Such

cracks could be found on PCB10 and PCB15 coatings as well (Figure 5H–J).

Table 2 summarizes the RMS values of the different coatings in MQ water and in SW. In general, the highest RMS roughness values with $23.7 \pm 1.5\ \text{nm}$ were found for the hydrophobic PDC coatings, while the hydrophilic PCB100 showed very low RMS roughness values of $1.9 \pm 0.4\ \text{nm}$ characteristic for smooth surfaces (Table 2). We were surprised that the incorporation of CBMA did not cause an increase in roughness but led to very smooth surfaces in PCB5, PCB10, and PCB15, with RMS values $\leq 4.5 \pm 0.3\ \text{nm}$. The RMS roughness in Table 2 shows no statistically significant differences between the amphiphilic coatings PCB5, PCB10, and PCB15 in the dry and in the wet state. On PDC, on the contrary, the saltwater conditions caused a noticeably higher surface roughness (RMS roughness values of $54.1 \pm 2.7\ \text{nm}$). Images of the lateral deflection of the cantilever (Figure S1) were used as a quantifiable analysis to determine the heterogeneity of the surface and gain insight into physiochemical differences due to the potential formation

Table 2. RMS Roughness, Thickness of the Polymer Coatings, and RMS of the Lateral Deflection of the Cantilever as Determined by AFM

	RMS roughness dry (nm)	RMS roughness wet (nm)	coating thickness (μm)	RMS lateral deflection (mV)
PDC	23.7 ± 1.5	54.1 ± 2.7	87.2 ± 0.6	3.7 ± 1.1
PCB5	4.5 ± 0.3	4.1 ± 0.6	88.7 ± 0.2	0.8 ± 0.3
PCB10	2.0 ± 0.8	2.1 ± 0.3	81.8 ± 0.3	0.8 ± 0.2
PCB15	2.3 ± 0.3	5.4 ± 0.9	85.2 ± 0.6	1.7 ± 0.2
PCB100	1.9 ± 0.4	1.3 ± 0.2	63.0 ± 0.7	0.4 ± 0.1

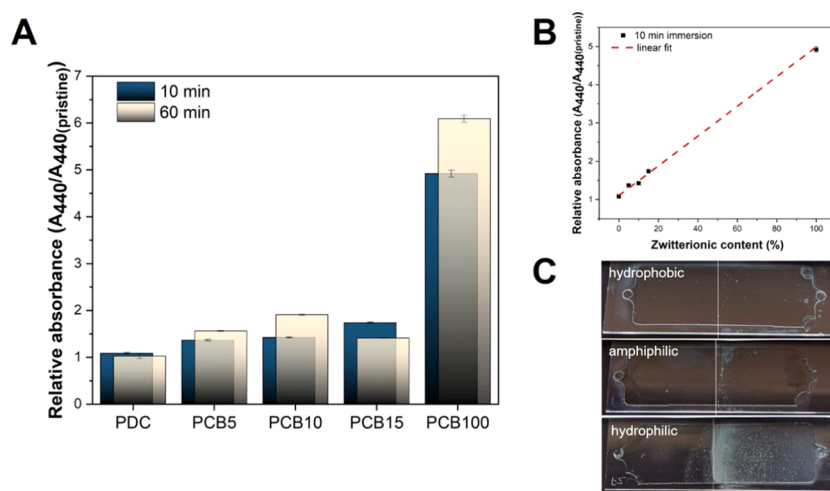


Figure 6. Increase of the UV–vis absorption of the polymer coatings after immersion in sand for 10 and 60 min, respectively. (A) Increase in the relative UV–vis absorbance of the immersed samples (10 and 60 min) in relation to the absorption of the pristine samples at 440 nm. Each value represents the average of three measurements. Error bars are obtained by the standard error of the mean. (B) Correlation between zwitterionic contents in the amphiphilic polymers and relative absorbance at 440 nm (sand uptake, values taken from (A) 10 min immersion). (C) Representative images of hydrophobic (PDC), amphiphilic (generic PCB5), and hydrophilic (PCB100) coatings as a pristine sample (left) and a sand-immersed sample (right).

of domains in the amphiphilic coatings. PDC showed the highest RMS values with 3.7 ± 1.1 , which corresponds to the higher roughness. The RMS values of the lateral deflection (Table 2) for PCB5 (0.8 ± 0.3 mV), PCB10 (0.8 ± 0.2 mV), and PCB15 (1.7 ± 0.2 mV) were very close to those of the hydrophilic PCB100 (0.4 ± 0.1 mV), which indicates that the coatings have a comparable homogeneity without visible domain formation in the lateral deflection images in Figure S1. The film thickness of the polymer depends on the height of the sealing chamber, the evaporation of the solvent, the cross-linking ratio, and the arrangement of monomer units in the polymer. PCD ($87 \mu\text{m}$) and the amphiphilic systems PCB5 ($89 \mu\text{m}$), PCB10 ($82 \mu\text{m}$), and PCB15 ($85 \mu\text{m}$) exhibit similar film thicknesses, while for PCB100, the thickness is only $63 \mu\text{m}$ even though the same sealing chamber was used.

In previous studies, we showed that zwitterionic hydrogels tend to incorporate inorganic substances and diatomaceous soil, also known as silt or marine snow, when exposed to the ocean.⁴² This uptake of sand altered the surfaces and changed the antifouling performance of the coating. To quantify the tendency of coatings to adsorb marine sediments, we developed a silt assay.⁴² Samples are exposed to sediment suspensions collected at the respective testing sites and autoclaved. After immersion of the coated samples into the suspensions on a laboratory shaking table for the desired time, the amount of adsorbed silt is determined by UV–vis absorption spectroscopy. Figure 6 shows the UV–vis absorption of polymer samples immersed in sand sediment, collected at the field-testing site in Port Canaveral (Florida), for 10 and 60 min. The UV–vis absorption of the immersed polymer samples at 440 nm was referenced to the absorption at 440 nm of the respective pristine samples. Even after this short immersion duration, the UV–vis data revealed a change in absorption, which directly correlated with the uptake of sand. As expected, the hydrophobic PCD polymer showed the lowest sediment absorption. When the zwitterionic CBMA was integrated into the coating, the absorption increased with increasing zwitterionic content. At only 5 wt % CBMA in the amphiphilic polymer PCB5, the uptake was already higher than that in the hydrophobic control PCD (Figure 6A). The by far highest sediment uptake was obtained for the pure zwitterionic polymer PCB100, which showed 5 times higher absorption than PCD within the 10 min immersion time. For the amphiphilic polymer coatings PCB5, PCB10, and PCB15, a steady increase of absorbed sand was observed with increasing hydrophilicity. When the immersion time was extended to 60 min, the observations followed the same trend, with the exception of PCB15, for which slightly lower absorption was detected. Interestingly, when compared to the 10 min measurement, the overall increase of absorbed sediment appeared to be rather marginal regarding the sixfold longer immersion time, which leads to the conclusion that silt uptake is a rather quick process. When the relative absorbance was plotted against the zwitterionic content of CBMA in the polymer coatings, a linear correlation (slope = 3.88, $R^2 = 0.98$) was observed (Figure 6B). Thus, the degree of sand incorporation is a function of hydrophilicity for the polymers consisting of DCPEA and CBMA.

Microfluidic Attachment Assay of the Marine Diatom *N. perminuta*. To evaluate the antifouling properties of the coatings, they were tested against the attachment of the diatom *N. perminuta* under dynamic conditions in ASW. The attachment assay was carried out in a custom-built micro-

fluidic setup, where the relative diatom attachment is known to reflect their attachment strength.⁵¹ As a hydrophobic control, butyl methacrylate (BMA) and an amphiphilic

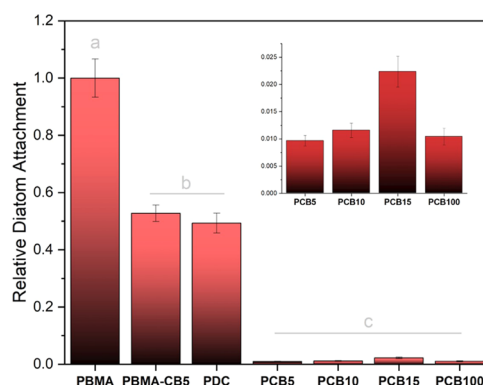


Figure 7. Diatom densities of *N. perminuta* on polymer coatings relative to negative control PBMA after a 90 min dynamic microfluidic assay with a 10 min rinsing step at a wall shear stress of 0.16 Pa. Each value is the average of three measurements with 30 fields of views per replicate. Error bars represent the standard error of the mean. Polymers are labeled a–c to show the statistical significance of the observed differences: polymers with different labels show significant differences, while polymers with identical labels do not show statistically significant differences.

polymer of the hydrophobic reference BMA combined with 5 wt % CBMA were included in the experiment. The highest accumulation of *N. perminuta* was observed on the hydrophobic PBMA (Figure 7). As experiments were performed on different days, all analyzed samples were normalized to the diatom densities on the PBMA coating. PCD (99°) shows similar hydrophobicity as PBMA (97°) but reduced the attachment by 50%, demonstrating that solely the wettability is not the only physiochemical property that determines cellular attachment. The incorporation of zwitterionic compounds into the coatings enhanced the antifouling performance with relative diatom attachment <0.03 for the amphiphilic polymer coatings (PCB5, PCB10, and PCB15). Even for the negative polymer control PBMA, the integration of 5 wt % CBMA decreased the relative diatom adhesion by nearly 50% (PBMA-CB5). While PCB5 showed the lowest accumulation, no further improvement was observed with higher content of the zwitterion. In fact, it rather seemed that even a slightly higher settlement occurred on PCB10 and PCB15 compared to PCB5, albeit this difference was not statistically significant. In Figure 7, the bars representing the diatom densities are labeled with the characters a–c. Polymers with different labels show significant differences, while polymers with identical labels are not statistically significantly different.

Rotating Disk Field Assay. Additional antifouling tests were done in a dynamic field assay using a rotating disk. The samples were immersed in the ocean for 5 days at a testing site at Port Canaveral in Florida to test the performance of the polymer coatings under dynamic conditions in a multispecies environment. The disk was operated at rotation speeds of 14 rpm to exert a similar wall shear stress of 0.18 Pa, which was also applied in the dynamic microfluidic laboratory experiments.⁵¹ Again, the hydrophobic control coating PBMA showed very high fluorescence coverage of nearly 23%, which was halved (11%) when the polymer was blended

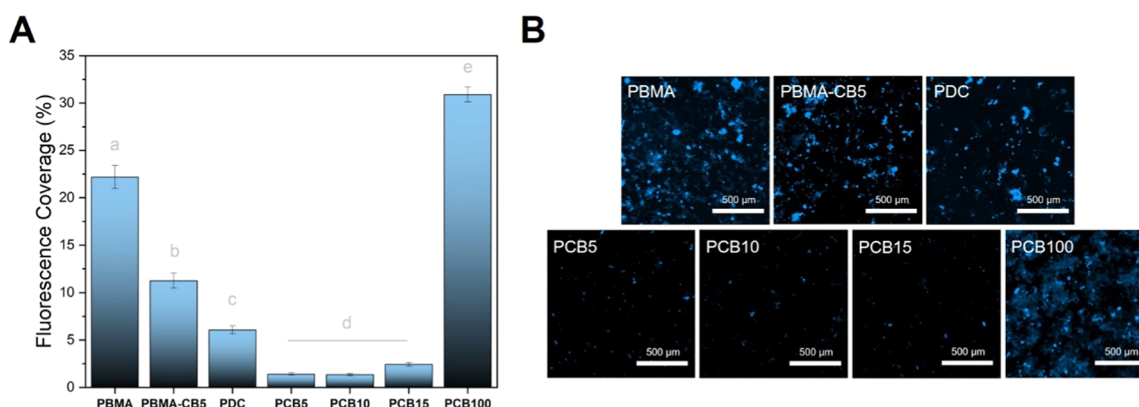


Figure 8. (A) Fluorescence coverage of polymer coatings tested in the field on a rotating disk at a rotation speed of 14 rpm submerged for 5 days at Port Canaveral, Florida. Each data point is the average of 3×60 fields of view recorded on the three independent sample replicates. The error bars represent the standard error of the mean. Polymers are labeled a–e to show the statistical significance of the observed differences: polymers with different labels show significant differences, while polymers with identical labels do not show statistically significant differences. (B) Representative fluorescence images of the polymer coatings.

with only 5 wt % CBMA (PBMA-CB5). Moreover, the PDC polymer consisting of the cyclic monomer DCPEA was found to reduce settlement to almost a quarter (6%) on the surface compared to the hydrophobic PBMA control. Also, for the amphiphilic DCPEA-based coatings, the antifouling performance was markedly enhanced by incorporating zwitterions into the polymer matrix. No significant improvement was achieved on increasing the CBMA content from 5 wt % over 10 to 15 wt %, which is in line with the laboratory results. Strikingly, the purely zwitterionic coating PCB100 exhibited the least fouling resistance with a coverage of over 30% and was, therefore, more than 35% higher than the coverage of the hydrophobic control PBMA. In both cases, BMA and DCPEA, the amphiphilic coatings PBMA-CB5 and PCB5 showed enhanced fouling resistance in direct comparison to the corresponding hydrophobic homopolymer coatings PBMA and PDC. For PBMA-CB5, a settlement reduction of 50% was reached, while for PCB5, the fouling was even reduced by 77%, with reference to the respective homopolymers. In Figure 8, the bars representing the organism density are labeled with the characters a–e. Polymers with different labels show significant differences, while polymers with identical labels are not statistically significantly different.

DISCUSSION

Amphiphilic monomers based on DCPEA and CBMA were copolymerized on silyl-methacrylate prefunctionalized substrates. The concentrations of the cross-linker (3 wt %) and photoinitiator (1 wt %) in the monomer solutions were kept low for all prepared polymers because too high rigidity due to enhanced cross-linking would hamper the ability of the polymer to reorganize and form stable hydration layers. This concept allows the preparation of coatings that are covalently attached to the surface and do not suffer delamination. The ATR-FTIR data showed that the zwitterionic component was successfully integrated into the hydrophobic matrix as a linear dependency between the zwitterionic content and the carboxyl/carbonyl ratio was observed that concurred with the signal intensities in the homopolymers. The increased content of CBMA was clearly reflected in the increase of characteristic carboxyl $\text{C}=\text{O}$ (1583 cm^{-1}) and OH-stretching vibration (3300 cm^{-1}) signals and in the decrease of the sessile static water contact angle in air, which shifted from

hydrophobic (99° in the case of PDC) to hydrophilic (42° for PCB100).

As the monomer composition and the amount and type of solvents are known to affect the morphology of coatings,⁴⁷ we studied the surface structure by AFM. The morphological analysis showed that increasing the CBMA content in the coating led to smoother surfaces, with RMS values between 27 nm (PDC) and $<5\text{ nm}$ (PCB5, PCB10, PCB15), in contrast to $<2\text{ nm}$ for the pure CBMA polymer (PCB100). Apart from the hydrophobic PDC, the obtained roughnesses were very similar and thus morphological influences on the results are very unlikely. This was surprising as we intuitively expected the potential formation of domains in the amphiphilic coatings, which would create enhanced roughness. The absence of phase segregation was shown by analysis of the lateral deflection image, which showed high chemical homogeneity for the amphiphilic surfaces. The higher roughness of the pure hydrophobic DCPEA films could be a solvent effect.⁴⁰ The observed relatively low and similar roughness for all of the prepared polymer coatings is helpful for their comparison as the surface geometry and topography are known to affect the antifouling properties.^{55–61}

When immersed in water, the coatings were stable for at least 13 days and the composition of the amphiphilic polymer as well as the mass of the coatings remained unchanged. No change in surface morphology was noticeable for the untreated, dry polymers and the ones immersed in saltwater for PCB5, PCB10, PCB15, or PCB100. The roughness of the PDC coatings did increase when immersed in SW though, which could have affected its antifouling properties. Also, we observed a rapid and sharp decrease of contact angles from air to water within minutes for PCB100 and for amphiphilic polymers with only low zwitterionic contents (PCB5, PCB10, and PCB15). This indicates a quick reorganization process under water resulting in a hydrophilic termination of the interface with contact angles $<20^\circ$ as determined by captive air bubble measurements.

The contact angle changes already indicated a reorganization of the surface for an optimized interaction with water. The amount of water bound by the coating was determined by gravimetry. It became apparent that PCB100 had a high capacity to absorb large amounts of water (244 wt % relative to the mass of the coatings), much higher than in the case of

the amphiphilic polymers PCB5 (3 wt %), PCB10 (7 wt %), PCB15 (6 wt %), and the hydrophobic control coating PDC (1 wt %). The amount of water absorbed by the coating is indicative of the hydrogel character of the coatings and the thickness of the hydrated part of the material. Obviously, a substantially larger fraction of the PCB100 coating is hydrated compared to the amphiphilic counterparts. Experimentally very difficult to assess, this could lead to a more diffuse interphase under water for PCB100, while the amphiphilic coatings most likely have a more compact hydrogel layer at the interface.

The large water absorption of PCB100 was connected to a high silt uptake from a silt suspension as determined by UV/vis spectroscopy, resulting in absorption that was more than threefold higher for PCB100 than that for PCB15 and fivefold higher than that for PDC samples after an immersion time of only 10 min. When the immersion time of the samples tested in the sand uptake assay was extended from 10 to 60 min, only a marginal increase of incorporated silt was observed in all polymer coatings, indicating that the silt uptake is a quick process. For the uptake of sand, a linear increase with zwitterionic content in the polymer was found. In summary, all zwitterion-containing coatings showed rapid reorganization under water, exposing the hydrophilic compounds into solution. We noted a clear correlation between the hydrophilic content, the swelling behavior, and the thickness of the hydrated interphase of the polymer coatings. A more diffuse interphase enabled the incorporation of silt.

For the investigated polyzwitterions, a remarkable reduction of fouling was observed in both the microfluidic attachment assay of the marine diatom *N. perminuta* and the dynamic field assay in a multiple-species environment when CBMA is incorporated into a hydrophobic matrix. Also, for PBMA, the antifouling capability improved twofold by integrating low contents of zwitterionic polymers (5 wt %). The findings revealed further that not only the choice and amount of hydrophilic material but, with equal importance, the choice of the hydrophobic compounds also plays a key role in the design of suitable amphiphilic polymer coatings for antifouling applications. The ethylene glycol dicyclopentenyl ether acrylate PDC showcased already impressive fouling resistance against pathogenic freshwater bacteria when copolymerized with several hydrophilic (meth)acrylate components.³² It was also found that the embedment of CBMA into BMA positively influenced the resistance to cell adhesion, with an optimum at a CBMA content of around 30 mol %.⁴³ Although the properties of cell culture media are different compared to seawater and a different hydrophobic monomer was used, the results are in accordance to the fouling results observed in our study. In our work, the accumulation of marine organisms on PDC was significantly lower (50%) than that on PBMA both in comparison to the homopolymers as well as when amphiphilic polymers with comparable CBMA content were compared. The advantages of amphiphilic polymers are supported by the good antifouling performance of PEGylated hyperbranched fluoropolymers on *Ulva*.¹⁸

The necessity of the combination of hydrophobic and hydrophilic units becomes particularly obvious when used in the field. As shown in several recent studies, experiments in the laboratory correlate frequently with short-term field exposure of multiple days in the real ocean environment but not in all.^{42,51} For example, hydrogel coatings with large water uptake and a diffuse interphase that exhibit ultralow fouling in

laboratory experiments suffer a functional loss in the field and are challenged by the incorporation of silt. The hydrogel character seems to enable the adsorption and incorporation of diatomaceous soil and other inorganic material, which causes a decay of the inertness of the coating as it provides an opportunity for marine fouling organisms to attach. This is in good agreement with the results of our field assay obtained for the purely zwitterionic polymer coating PCB100. While the microfluidic attachment assay with *N. perminuta* delivered good fouling resistance, the short-term exposure to the real marine environment in the rotating disk assay led to a major accumulation of marine organisms, yielding a fluorescence coverage of 30%. This was a high value, even compared to the hydrophobic negative control PBMA exhibiting a fluorescence coverage of 22% in the field and the highest diatom attachment in the lab assays.

As a conclusion, we note that in laboratory experiments, hydrophilic termination is essential for good antifouling results. In the field, however, the strong swelling of the zwitterionic polymers and the associated diffuse interphase is prone to the incorporation of silt, which hampers the low-fouling properties. Hence, a suitable hydrophilic termination, without generating a too diffuse interphase to reduce silt uptake, seems to be a suitable design rule for optimized performance in the field. In both dynamic microfluidic attachment assays as well as in the dynamic rotating disk field assay, the amphiphilic polymers PCB5, PCB10, and PCB15 exhibited consistently ultralow fouling properties and showed strong enhancement to repel the attachment of microorganisms. The good performance even with a low zwitterionic content embedded in a hydrophobic matrix implies that polymers with moderate hydrophilicity and moderate swelling still allowed the zwitterionic compound to unfold its low fouling properties while reducing the silt uptake. This optimal combination of hydrophilic and hydrophobic moieties in amphiphilic polymers seems to be perfectly suited to overcome both the fouling and the silt incorporation challenges in the field.

CONCLUSIONS

A set of amphiphilic polymer coatings containing various amounts of carboxy-terminated zwitterions and dicyclopentadienyl as hydrophobic ingredients were synthesized utilizing photoinduced free-radical polymerization. This polymerization technique under controlled oxygen-free conditions allowed us to mix acrylic and methacrylic monomers with different hydrophilicities using appropriate solvents and in situ copolymerization on chemically prefunctionalized surfaces. The stability results proved that this is a suitable platform for a quick and reliable preparation of polymers for antifouling material screens. The resulting coatings were characterized by AFM, water contact angle goniometry, and ATR-FTIR. The polymers showed very good antifouling behavior against *N. perminuta* as determined in the dynamic microfluidic accumulation assay and in the dynamic rotating disk assay in the field. We demonstrated that zwitterions can be applied as antifouling compounds in the field that even withstand silt incorporation into the polymers if sufficient amphiphilicity is accomplished. Notably, the choice of the hydrophobic base is a key factor and the use of DCPEA shows significant advantages over the use of butyl methacrylate.

■ ASSOCIATED CONTENT

■ Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsami.0c07599>.

Representative lateral deflection images in SW by AFM; comparison of the statistical significance between polymer coatings in the dynamic attachment assay with *N. perminuta*; and comparison of the statistical significance between polymer coatings in the rotating disk field assay (PDF)

■ AUTHOR INFORMATION

Corresponding Author

Axel Rosenhahn – Analytical Chemistry—Biointerfaces, Ruhr University Bochum, 44780 Bochum, Germany; orcid.org/0000-0001-9393-7190; Email: axel.rosenhahn@rub.de

Authors

Florian Koschitzki – Analytical Chemistry—Biointerfaces, Ruhr University Bochum, 44780 Bochum, Germany; orcid.org/0000-0002-0511-0431

Robin Wanka – Analytical Chemistry—Biointerfaces, Ruhr University Bochum, 44780 Bochum, Germany; orcid.org/0000-0001-8892-9399

Lennart Sobota – Analytical Chemistry—Biointerfaces, Ruhr University Bochum, 44780 Bochum, Germany

Julian Koc – Analytical Chemistry—Biointerfaces, Ruhr University Bochum, 44780 Bochum, Germany; orcid.org/0000-0001-6026-9368

Harrison Gardner – Center for Corrosion and Biofouling Control, Florida Institute of Technology, Melbourne, Florida 32901, United States

Kelli Z. Hunsucker – Center for Corrosion and Biofouling Control, Florida Institute of Technology, Melbourne, Florida 32901, United States

Geoffrey W. Swain – Center for Corrosion and Biofouling Control, Florida Institute of Technology, Melbourne, Florida 32901, United States

Complete contact information is available at: <https://pubs.acs.org/doi/10.1021/acsami.0c07599>

Funding

Funding: ONR N00014-16-12979, N00014-16-13123 and N0014-20-12244, DFG RO 2524/4-1, and DFG 2524/4-2.

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

We acknowledge helpful discussions with Prof. A. Laschewsky, University Potsdam, and Prof. M. Alexander, University of Nottingham. The authors further acknowledge access to the atomic force microscopy by Prof. W. Schuhmann, Analytical Chemistry-Electroanalytics and Sensing, RUB.

■ REFERENCES

- (1) Townsin, R. L. The Ship Hull Fouling Penalty. *Biofouling* **2003**, *19*, 9–15.
- (2) Schultz, M. P.; Bendick, J. A.; Holm, E. R.; Hertel, W. M. Economic Impact of Biofouling on a Naval Surface Ship. *Biofouling* **2011**, *27*, 87–98.

- (3) Davidson, I. C.; Brown, C. W.; Sytsma, M. D.; Ruiz, G. M. The Role of Containerships as Transfer Mechanisms of Marine Biofouling Species. *Biofouling* **2009**, *25*, 645–655.

- (4) Johnson, L. E.; Ricciardi, A.; Carlton, J. T. Overland Dispersal of Aquatic Invasive Species: A Risk Assessment of Transient Recreational Boating. *Ecol. Appl.* **2001**, *11*, 1789–1799.

- (5) Lejars, M.; Margaillan, A.; Bressy, C. Fouling Release Coatings: A Nontoxic Alternative to Biocidal Antifouling Coatings. *Chem. Rev.* **2012**, *112*, 4347–4390.

- (6) Magin, C. M.; Cooper, S. P.; Brennan, A. B. Non-Toxic Antifouling Strategies. *Mater. Today* **2010**, *13*, 36–44.

- (7) Kirschner, C. M.; Brennan, A. B. Bio-Inspired Antifouling Strategies. *Annu. Rev. Mater. Res.* **2012**, *42*, 211–229.

- (8) Baier, R. E. Surface Behaviour of Biomaterials: The Theta Surface for Biocompatibility. *J. Mater. Sci.: Mater. Electron.* **2006**, *17*, 1057–1062.

- (9) Rosenhahn, A.; Schilp, S.; Kreuzer, H. J.; Grunze, M. The Role of “Inert” Surface Chemistry in Marine Biofouling Prevention. *Phys. Chem. Chem. Phys.* **2010**, *12*, 4275–4286.

- (10) Higaki, Y.; Nishida, J.; Takenaka, A.; Yoshimatsu, R.; Kobayashi, M.; Takahara, A. Versatile Inhibition of Marine Organism Settlement by Zwitterionic Polymer Brushes. *Polym. J.* **2015**, *47*, 811–818.

- (11) Jakobi, V.; Schwarze, J.; Finlay, J. A.; Nolte, K. A.; Spölmann, S.; Becker, H. W.; Clare, A. S.; Rosenhahn, A. Amphiphilic Alginates for Marine Antifouling Applications. *Biomacromolecules* **2018**, *19*, 402–408.

- (12) Callow, J. A.; Callow, M. E. Trends in the Development of Environmentally Friendly Fouling-Resistant Marine Coatings. *Nat. Commun.* **2011**, *2*, No. 244.

- (13) Krishnan, S.; Weinman, C. J.; Ober, C. K. Advances in Polymers for Anti-Biofouling Surfaces. *J. Mater. Chem.* **2008**, *18*, 3405–3413.

- (14) Galli, G.; Martinelli, E. Amphiphilic Polymer Platforms: Surface Engineering of Films for Marine Antibiofouling. *Macromol. Rapid Commun.* **2017**, *38*, No. 1600704.

- (15) Krishnan, S.; Ayothi, R.; Hexemer, A.; Finlay, J. A.; Sohn, K. E.; Perry, R.; Ober, C. K.; Kramer, E. J.; Callow, M. E.; Callow, J. A.; Fischer, D. A. Anti-Biofouling Properties of Comblike Block Copolymers with Amphiphilic Side Chains. *Langmuir* **2006**, *22*, 5075–5086.

- (16) Martinelli, E.; Sarvothaman, M. K.; Galli, G.; Pettitt, M. E.; Callow, M. E.; Callow, J. A.; Conlan, S. L.; Clare, A. S.; Sugiharto, A. B.; Davies, C.; Williams, D. Poly(Dimethyl Siloxane) (PDMS) Network Blends of Amphiphilic Acrylic Copolymers with Poly-(Ethylene Glycol)-Fluoroalkyl Side Chains for Fouling-Release Coatings. II. Laboratory Assays and Field Immersion Trials. *Biofouling* **2012**, *28*, 571–582.

- (17) Gudipati, C. S.; Greenleaf, C. M.; Johnson, J. A.; Prayoncan, P.; Wooley, K. L. Hyperbranched Fluoropolymer and Linear Poly(Ethylene Glycol) Based Amphiphilic Crosslinked Networks as Efficient Antifouling Coatings: An Insight into the Surface Compositions, Topographies, and Morphologies. *J. Polym. Sci., Part A: Polym. Chem.* **2004**, *42*, 6193–6208.

- (18) Gudipati, C. S.; Finlay, J. A.; Callow, J. A.; Callow, M. E.; Wooley, K. L. The Antifouling and Fouling-Release Performance of Hyperbranched Fluoropolymer (HBFP)-Poly(Ethylene Glycol) (PEG) Composite Coatings Evaluated by Adsorption of Biomacromolecules and the Green Fouling Alga *Ulva*. *Langmuir* **2005**, *21*, 3044–3053.

- (19) Martinelli, E.; Agostini, S.; Galli, G.; Chiellini, E.; Glisenti, A.; Pettitt, M. E.; Callow, M. E.; Callow, J. A.; Graf, K.; Bartels, F. W. Nanostructured Films of Amphiphilic Fluorinated Block Copolymers for Fouling Release Application. *Langmuir* **2008**, *24*, 13138–13147.

- (20) Bauer, S.; Alles, M.; Arpa-Sancet, M. P.; Ralston, E.; Swain, G. W.; Aldred, N.; Clare, A. S.; Finlay, J. A.; Callow, M. E.; Callow, J. A.; Rosenhahn, A. Resistance of Amphiphilic Polysaccharides against Marine Fouling Organisms. *Biomacromolecules* **2016**, *17*, 897–904.

- (21) Bodkhe, R. B.; Stafslie, S. J.; Daniels, J.; Cilz, N.; Muelhberg, A. J.; Thompson, S. E. M.; Callow, M. E.; Callow, J. A.; Webster, D. C. Zwitterionic Siloxane-Polyurethane Fouling-Release Coatings. *Prog. Org. Coat.* **2015**, *78*, 369–380.
- (22) Chen, S.; Jiang, S. A New Avenue to Nonfouling Materials. *Adv. Mater.* **2008**, *20*, 335–338.
- (23) Shahkaramipour, N.; Ramanan, S. N.; Fister, D.; Park, E.; Venna, S. R.; Sun, H.; Cheng, C.; Lin, H. Facile Grafting of Zwitterions onto the Membrane Surface to Enhance Antifouling Properties for Wastewater Reuse. *Ind. Eng. Chem. Res.* **2017**, *56*, 9202–9212.
- (24) Holmlin, R. E.; Chen, X.; Chapman, R. G.; Takayama, S.; Whitesides, G. M. Zwitterionic SAMs That Resist Nonspecific Adsorption of Protein from Aqueous Buffer. *Langmuir* **2001**, *17*, 2841–2850.
- (25) Leng, C.; Hung, H. C.; Sun, S.; Wang, D.; Li, Y.; Jiang, S.; Chen, Z. Probing the Surface Hydration of Nonfouling Zwitterionic and PEG Materials in Contact with Proteins. *ACS Appl. Mater. Interfaces* **2015**, *7*, 16881–16888.
- (26) Schilp, S.; Kueller, A.; Rosenhahn, A.; Grunze, M.; Pettitt, M. E.; Callow, M. E.; Callow, J. A. Settlement and Adhesion of Algal Cells to Hexa(Ethylene Glycol)-Containing Self-Assembled Monolayers with Systematically Changed Wetting Properties. *Biointerphases* **2007**, *2*, 143–150.
- (27) Cheng, G.; Li, G.; Xue, H.; Chen, S.; Bryers, J. D.; Jiang, S. Zwitterionic Carboxybetaine Polymer Surfaces and Their Resistance to Long-Term Biofilm Formation. *Biomaterials* **2009**, *30*, 5234–5240.
- (28) Aldred, N.; Li, G.; Gao, Y.; Clare, A. S.; Jiang, S. Modulation of Barnacle (*Balanus Amphitrite* Darwin) Cyprid Settlement Behavior by Sulfobetaine and Carboxybetaine Methacrylate Polymer Coatings. *Biofouling* **2010**, *26*, 673–683.
- (29) Bauer, S.; Alles, M.; Finlay, J. A.; Callow, J. A.; Callow, M. E.; Rosenhahn, A. Influence of Zwitterionic SAMs on Protein Adsorption and the Attachment of Algal Cells. *J. Biomater. Sci., Polym. Ed.* **2014**, *25*, 1530–1539.
- (30) Jiang, S.; Ishihara, K.; Ji, J. Special Issue on Zwitterionic Materials. *Acta Biomater.* **2016**, *40*, iv.
- (31) Laschewsky, A.; Rosenhahn, A. Molecular Design of Zwitterionic Polymer Interfaces: Searching for the Difference. *Langmuir* **2019**, *35*, 1056–1071.
- (32) Hook, A. L.; Chang, C. Y.; Yang, J.; Luckett, J.; Cockayne, A.; Atkinson, S.; Mei, Y.; Bayston, R.; Irvine, D. J.; Langer, R.; Anderson, D. G.; Williams, P.; Davies, M. C.; Alexander, M. R. Combinatorial Discovery of Polymers Resistant to Bacterial Attachment. *Nat. Biotechnol.* **2012**, *30*, 868–875.
- (33) Hayward, J. A.; Chapman, D. Biomembrane Surfaces as Models for Polymer Design: The Potential for Haemocompatibility. *Biomaterials* **1984**, *5*, 135–142.
- (34) Ishihara, K.; Aragaki, R.; Ueda, T.; Watanabe, A.; Nakabayashi, N. Reduced Thrombogenicity of Polymers Having Phospholipid Polar Groups. *J. Biomed. Mater. Res.* **1990**, *24*, 1069–1077.
- (35) Zhang, Z.; Chao, T.; Chen, S.; Jiang, S. Superlow Fouling Sulfobetaine and Carboxybetaine Polymers on Glass Slides. *Langmuir* **2006**, *22*, 10072–10077.
- (36) Vaisocherová, H.; Yang, W.; Zhang, Z.; Cao, Z.; Cheng, G.; Piliarik, M.; Homola, J.; Jiang, S. Ultralow Fouling and Functionalizable Surface Chemistry Based on a Zwitterionic Polymer Enabling Sensitive and Specific Protein Detection in Undiluted Blood Plasma. *Anal. Chem.* **2008**, *80*, 7894–7901.
- (37) Li, G.; Xue, H.; Cheng, G.; Chen, S.; Zhang, F.; Jiang, S. Ultralow Fouling Zwitterionic Polymers Grafted from Surfaces Covered with an Initiator via an Adhesive Mussel Mimetic Linkage. *J. Phys. Chem. B* **2008**, *112*, 15269–15274.
- (38) Zhang, Z.; Finlay, J. A.; Wang, L.; Gao, Y.; Callow, J. A.; Callow, M. E.; Jiang, S. Polysulfobetaine-Grafted Surfaces as Environmentally Benign Ultralow Fouling Marine Coatings. *Langmuir* **2009**, *25*, 13516–13521.
- (39) Mahmud, G.; Huda, S.; Yang, W.; Kandere-Grzybowska, K.; Pilans, D.; Jiang, S.; Grzybowski, B. A. Carboxybetaine Methacrylate Polymers Offer Robust, Long-Term Protection against Cell Adhesion. *Langmuir* **2011**, *27*, 10800–10804.
- (40) Koc, J.; Schönmann, E.; Amuthalingam, A.; Clarke, J.; Finlay, J. A.; Clare, A. S.; Laschewsky, A.; Rosenhahn, A. Low Fouling Thin Hydrogel Coatings Made of Photo-Crosslinked Polyzwitterions. *Langmuir* **2019**, *35*, 1552–1562.
- (41) Schönmann, E.; Laschewsky, A.; Wischerhoff, E.; Koc, J.; Rosenhahn, A. Surface Modification by Polyzwitterions of the Sulfobetaine-Type, and Their Resistance to Biofouling. *Polymers* **2019**, *11*, 1014–1046.
- (42) Koc, J.; Simovich, T.; Schönmann, E.; Chilkoti, A.; Gardner, H.; Swain, G. W.; Hunsucker, K.; Laschewsky, A.; Rosenhahn, A. Sediment Challenge to Promising Ultra-Low Fouling Hydrophilic Surfaces in the Marine Environment. *Biofouling* **2019**, *35*, 454–462.
- (43) Lin, X.; Jain, P.; Wu, K.; Hong, D.; Hung, H. C.; O'Kelly, M. B.; Li, B.; Zhang, P.; Yuan, Z.; Jiang, S. Ultralow Fouling and Functionalizable Surface Chemistry Based on Zwitterionic Carboxybetaine Random Copolymers. *Langmuir* **2019**, *35*, 1544–1551.
- (44) Ventura, C.; Guerin, A. J.; El-Zubir, O.; Ruiz-Sanchez, A. J.; Dixon, L. I.; Reynolds, K. J.; Dale, M. L.; Ferguson, J.; Houlton, A.; Horrocks, B. R.; Clare, A. S.; Fulton, D. A. Marine Antifouling Performance of Polymer Coatings Incorporating Zwitterions. *Biofouling* **2017**, *33*, 892–903.
- (45) Thome, I.; Bauer, S.; Vater, S.; Zargiel, K.; Finlay, J. A.; Arpa-Sancet, M. P.; Alles, M.; Callow, M. E.; Swain, G. W.; Grunze, M.; Rosenhahn, A. Conditioning of Self-Assembled Monolayers at Two Static Immersion Test Sites along the East Coast of Florida and Its Effect on Early Fouling Development. *Biofouling* **2014**, *30*, 1011–1021.
- (46) Liu, Z.; Zhang, B.; Pow, B. Y. F.; Yang, M.; Mak, A. F. T. Construction of Nonbiofouling Biofunctional Glass Surface by Self-Assembled Monolayer and Graft Hydrophilic Polymer. *Adv. Mater. Res.* **2008**, *47–50*, 1343–1346.
- (47) Keller, N.; Bruchmann, J.; Sollich, T.; Richter, C.; Thelen, R.; Kotz, F.; Schwartz, T.; Helmer, D.; Rapp, B. E. Study of Biofilm Growth on Slippery Liquid-Infused Porous Surfaces Made from Fluoropor. *ACS Appl. Mater. Interfaces* **2019**, *11*, 4480–4487.
- (48) Kester, D. R.; Duedall, I. W.; Conners, D. N.; Pytkowicz, R. M. Department of Zoology and Plankton Laboratory, University of Southampton, England. *Limnol. Oceanogr.* **1967**, *176*–179.
- (49) Thome, I.; Pettitt, M. E.; Callow, M. E.; Callow, J. A.; Grunze, M.; Rosenhahn, A. Conditioning of Surfaces by Macromolecules and Its Implication for the Settlement of Zoospores of the Green Alga *Ulva linza*. *Biofouling* **2012**, *28*, 501–510.
- (50) Nolte, K. A.; Schwarze, J.; Beyer, C. D.; Özcan, O.; Rosenhahn, A. Parallelized Microfluidic Diatom Accumulation Assay to Test Fouling-Release Coatings. *Biointerphases* **2018**, *13*, No. 041007.
- (51) Nolte, K. A.; Koc, J.; Barros, J. M.; Hunsucker, K.; Schultz, M. P.; Swain, G. E.; Rosenhahn, A. Dynamic Field Testing of Coating Chemistry Candidates by a Rotating Disk System. *Biofouling* **2018**, *34*, 398–409.
- (52) Shao, Q.; Jiang, S. Influence of Charged Groups on the Properties of Zwitterionic Moieties: A Molecular Simulation Study. *J. Phys. Chem. B* **2014**, *118*, 7630–7637.
- (53) Kudaibergenov, S.; Jaeger, W.; Laschewsky, A. Polymeric Betaines: Synthesis, Characterization, and Application. In *Supramolecular Polymers Polymeric Betains Oligomers*; Abe, A. et al., Eds.; Advances in Polymer Science; Springer-Verlag: Berlin, Heidelberg, 2006; Vol. 201, pp 157–224.
- (54) Wang, F.; Yang, J.; Zhao, J. Understanding Anti-Polyelectrolyte Behavior of a Well-Defined Polyzwitterion at the Single-Chain Level. *Polym. Int.* **2015**, *64*, 999–1005.
- (55) Scardino, A. J.; Harvey, E.; De Nys, R. Testing Attachment Point Theory: Diatom Attachment on Microtextured Polyimide Biomimics. *Biofouling* **2006**, *22*, 55–60.
- (56) Scardino, A. J.; Guenther, J.; de Nys, R. Attachment Point Theory Revisited: The Fouling Response to a Microtextured Matrix. *Biofouling* **2008**, *24*, 45–53.

(57) Cao, X.; Pettitt, M. E.; Wode, F.; Arpa Sancet, M. P.; Fu, J.; Ji, J.; Callow, M. E.; Callow, J. A.; Rosenhahn, A.; Grunze, M. Interaction of Zoospores of the Green Alga *Ulva* with Bioinspired Micro- and Nanostructured Surfaces Prepared by Polyelectrolyte Layer-by-Layer Self-Assembly. *Adv. Funct. Mater.* **2010**, *20*, 1984–1993.

(58) Carman, M. L.; Estes, T. G.; Feinberg, A. W.; Schumacher, J. F.; Wilkerson, W.; Wilson, L. H.; Callow, M. E.; Callow, J. A.; Brennan, A. B. Engineered Antifouling Microtopographies - Correlating Wettability with Cell Attachment. *Biofouling* **2006**, *22*, 11–21.

(59) Schumacher, J. F.; Carman, M. L.; Estes, T. G.; Feinberg, A. W.; Wilson, L. H.; Callow, M. E.; Callow, J. A.; Finlay, J. A.; Anthony, B. Engineered Antifouling Microtopographies – Effect of Feature Size, Geometry, and Roughness on Settlement of Zoospores of The green Alga *Ulva*. *Biofouling* **2007**, *23*, 55–62.

(60) Schumacher, J. F.; Aldred, N.; Callow, M. E.; Finlay, J. A.; Callow, J. A.; Clare, A. S.; Brennan, A. B. Species-Specific Engineered Antifouling Topographies: Correlations between the Settlement of Algal Zoospores and Barnacle Cyprids. *Biofouling* **2007**, *23*, 307–317.

(61) Schumacher, J. F.; Long, C. J.; Callow, M. E.; Finlay, J. A.; Callow, J. A.; Brennan, A. B. Engineered Nanoforce Gradients for Inhibition of Settlement (Attachment) of Swimming Algal Spores. *Langmuir* **2008**, *24*, 4931–4937.